

ROLLING MILL MOTOR MAINTENANCE MANUAL **Document Number:** RMM-MNT-001

Equipment Name: Rolling Mill Main Drive Motor

Equipment Type: Three-Phase Induction Motor

Application: Steel Rolling Mill

Revision: 1.0

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1. PURPOSE

This maintenance manual provides procedures, schedules, inspection guidelines, troubleshooting instructions, and safety requirements for maintaining the Rolling Mill Main Drive Motor.

2. EQUIPMENT DESCRIPTION

Rated Power: 2500 kW

Rated Voltage: 6600 V

Frequency: 50 Hz

Rated Speed: 1480 RPM

Motor Type: Squirrel Cage Induction Motor

Cooling Method: Forced Air Cooling

Protection Class: IP55

3. SAFETY REQUIREMENTS

LOTO: Shut down motor, isolate power supply, lock isolation switch, attach warning tag, verify zero voltage.

Required PPE: Safety Helmet, Electrical Gloves, Safety Shoes, Eye Protection, Hearing Protection.

4. DAILY INSPECTION CHECKLIST

Check oil leaks, abnormal vibrations, unusual noise, cooling fan condition.

Temperature: 50–80°C, Vibration: 0–2 mm/s, Current: 10–25 A, RPM: 1450–1500.

5. WEEKLY MAINTENANCE

Inspect lubrication, coupling condition, cable terminations, loose connections, junction box.

6. MONTHLY MAINTENANCE

Vibration Analysis: Acceptable ≤ 2 mm/s, Warning 2–3 mm/s, Critical > 3 mm/s.

Thermal inspection using infrared camera.

7. QUARTERLY MAINTENANCE

Alignment ≤ 0.05 mm. Insulation Resistance > 100 M Ω . Inspect bearings.

8. ANNUAL OVERHAUL

Bearing replacement, rotor inspection, stator inspection, shaft inspection, dynamic balancing, cooling system overhaul.

9. COMMON FAILURE MODES

Bearing Wear, Shaft Misalignment, Rotor Imbalance, Cooling System Failure.

10. TROUBLESHOOTING GUIDE

Motor Overheating, Excessive Vibration, High Current Draw investigation procedures.

11. ALARM CONDITIONS

High Temperature $> 90^\circ\text{C}$, Critical Temperature $> 100^\circ\text{C}$, High Vibration > 3 mm/s, Overcurrent > 30 A.

12. SPARE PARTS

B6205 Bearing, FAN-100 Cooling Fan, CPL-200 Coupling, SHAFT-10 Shaft, GREASE-L1.

13. ROOT CAUSE REFERENCE TABLE

High Temperature + High Vibration → Bearing Wear, etc.

14. MAINTENANCE DECISION MATRIX

Temperature >95°C → Immediate Shutdown.

Vibration >3 mm/s → Maintenance within 24 hours.

Current >30 A and RPM <1400 → Mechanical Overload.

15. MAINTENANCE RECORD REQUIREMENTS

Record date, equipment ID, technician, root cause, actions, downtime, spare parts, verification results.